Patrick Thangarajah

Simulation & FEA | Testing & Validation | Mechanical Design | Prototyping

Calgary, AB | 403-585-1903 | ptemmanual@gmail.com | www.linkedin.com/in/patrickthangarajah/ | TN eligible | Open to Relocation

Skills

Simulation & Programming	FEA, LS-DYNA, LS-PrePost, Python, C/C++
Model Validation and Testing	FE model validation, physical test correlation, ISO 18571, experimental data analysis
Design & Prototyping	SolidWorks, Onshape, Engineering Drawings, 3D Printing, Lathe, Mill, Bandsaw, Drill Press
Testing & Analysis	DOE, JMP, Minitab, MATLAB, Microsoft Excel, Troubleshooting
Technical Communication	Scientific literature reviews, technical reports, technical presentations, work instructions, SOPs

Education

Master of Applied Science in Mechanical and Mechatronics Engineering — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Research in impact mechanics, finite element analysis, and model validation
- Graduate coursework average: 90%

Bachelor of Applied Science in Mechanical Engineering, Co-op Program — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*
- Biomechanics and Medical Devices Option

Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Validated a THOR-5F ATD FE model against physical sled testing in two bench configurations in LS-DYNA, achieving an average ISO 18571 score of 0.71 across 10 measurement channels
- Designed and analyzed a 104-simulation FE parametric study of seat belt anchor locations using JMP DOE, least-squares regression, and ANOVA; identified a rearward and inboard position with 16% lower peak chest deflection than the study average
- Troubleshot model-to-test discrepancies in positioning, restraint fit, and response timing to improve agreement with physical tests
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)
- Presented research findings through a master's thesis, conference paper, technical reports and six sponsor presentations, including to Transport Canada

Teaching Assistant - MTE 120: Circuits — *Waterloo, ON, Canada*
University of Waterloo — *May 2024 - Aug 2024*
- Supported laboratory instruction for a 100+ student circuits course, troubleshooting hardware and providing feedback on lab work and exams

R&D Engineer (Part-time) — *Remote*
Echoscope — *May 2023 - Sep 2023*
- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functionality testing; all five were approved and used by the client

Summer Research Student — *Vancouver, BC, Canada (Remote)*
Centre for Heart Lung Innovation — *May 2022 - Aug 2022*
- Built a transfer-learning diagnostic model in a Jupyter Notebook, improving diabetic kidney disease detection accuracy by 10% through optimization, training refinement and tissue-slide image comparisons

Mechanical Engineering Intern — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*
- Created standardized work instructions and SOPs verified against engineering drawings and customer specifications
- Reduced fiberglass cutting waste by 50% through process analysis, and proposed a material handling workflow projected to halve pick-and-place time

Projects

Effect of Seat Foam Properties on Injury Response in Rear Impact (ME 725 Course Project) — *Waterloo, ON, Canada*
University of Waterloo — *Jan 2024 - Apr 2024*
- Verified 5 polyurethane foam material models in LS-DYNA against input stress-strain response using single-element simulations, then implemented them in a rear-impact sled model with a Hybrid III ATD to calculate HIC15 and Nij injury metrics
- Determined that all 5 foams met the regulatory head injury limit but exceeded the regulatory neck injury limit at high speed, and that the softest foam reduced head injury risk by roughly 18–20% relative to the stiffest

Soft Distensible Bladder Model - Capstone Project — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2022 - Apr 2023*
- Designed and built the control and data-acquisition interface for a pressurized soft-bladder test system, enabling real-time pressure/volume monitoring and automated filling/voiding cycles
- Validated system performance using integrated pressure sensing and controls, demonstrating stable and repeatable operation up to 4.75 kPa internal pressure

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684